

PHP Basic to Advanced Notes

PHP BASIC TO ADVANCED NOTES

1. Introduction to PHP

PHP = Hypertext Preprocessor. Server-side scripting language used for web development.

2. Variables

Variables start with \$.

Example: `$name = "Malik";`

3. Data Types

String, Integer, Float, Boolean, Array, Object, NULL.

4. Constants

`define("SITE","StudyHub");`

5. Operators

Arithmetic, Comparison, Logical, Assignment.

6. Input & Output

`echo`, `print`, `$_GET`, `$_POST`.

7. Conditional Statements

`if`, `if-else`, `if-elseif-else`, `switch`.

8. Loops

`for`, `while`, `do-while`, `foreach`.

9. Arrays

Indexed Array, Associative Array, Multidimensional Array.

10. Strings

`strlen()`, `str_replace()`, `strtoupper()`, `strtolower()`.

11. Functions

User-defined functions and built-in functions.

12. Superglobals

`$_GET`, `$_POST`, `$_REQUEST`, `$_SESSION`, `$_COOKIE`, `$_SERVER`.

13. Form Handling

Receiving user input through HTML forms.

14. Sessions

`session_start()`;

Used for storing user login data.

15. Cookies

setcookie() used to store data in browser.

16. File Handling

fopen(), fread(), fwrite(), fclose().

17. OOP in PHP

Class, Object, Constructor, Destructor.

18. Inheritance

Child class inherits parent class properties and methods.

19. Polymorphism

Same method with different behavior.

20. Encapsulation

Private and protected properties.

21. Abstraction

Abstract classes and methods.

22. Interfaces

Defines method contracts.

23. Exception Handling

try, catch, finally, throw.

24. MySQL Connection

mysqli_connect() or PDO.

25. CRUD Operations

Create, Read, Update, Delete.

26. PHP with HTML

Embed PHP inside HTML pages.

27. Authentication System

Login, Signup, Session Management.

28. Security

Password Hashing, SQL Injection Prevention, Validation.

29. PDO

Secure database operations using prepared statements.

30. PHP Frameworks

Laravel, CodeIgniter, Symfony.

Important Interview Topics:

Variables, Arrays, Functions, OOP, Sessions, Cookies, CRUD, MySQL, PDO, Security.